

Free IT & Cybersecurity Learning Cheat Sheet

2026 Edition • by Kevin Zelman

Every link below is free (or has a solid free tier). Bookmark it, pick one thing from each section, and start.
kevinzelman.github.io/mentoring.html • discord.gg/WJn4bQV9m9

Start here: maps of the field

- **roadmap.sh/cyber-security**: a visual, step-by-step skills roadmap.
- **CyberSeek.org**: security job titles, the skills they need, and common career pathways.
- **NICE Workforce Framework** (niccs.cisa.gov): the official U.S. list of security work roles and tasks.
- **My Cybersecurity Career Starter Guide**: paths, certifications and a 90-day plan (on my mentoring page).

Networking & operating systems

- **Professor Messer** (prolessormesser.com): free A+, Network+ and Security+ video courses.
- **Cisco Networking Academy** (netacad.com): free intro courses in networking and cybersecurity.
- **Jeremy's IT Lab** (YouTube): a complete free CCNA course. Great for networking fundamentals.
- **Linux Journey** (linuxjourney.com): learn Linux from zero, one short lesson at a time.
- **OverTheWire: Bandit** (overthewire.org): learn the Linux command line by playing a game.

Scripting & programming

- **Automate the Boring Stuff with Python** (automatetheboringstuff.com): free to read online, and practical from page one.
- **Harvard CS50** (cs50.harvard.edu): the best free intro to computer science.
- **freeCodeCamp** (freecodecamp.org): free courses in Python, JavaScript and more.
- **Microsoft Learn: PowerShell** (learn.microsoft.com/powershell): the language that runs Windows and Microsoft 365 admin.

Cloud, identity & Microsoft 365

- **Microsoft Learn** (learn.microsoft.com): free learning paths and hands-on sandboxes for AZ-900, SC-900, SC-300, AZ-104 and more.
- **Azure free account** (azure.microsoft.com/free): build a real Entra ID and Azure lab.
- **AWS Skill Builder** (skillbuilder.aws): free AWS Cloud Practitioner training.

Blue team: defend & detect

- **MITRE ATT&CK** (attack.mitre.org): how real attackers operate. It's the common language of detection.
- **Blue Team Labs Online** (blueteamlabs.online): investigation challenges using real-looking evidence.
- **CyberDefenders** (cyberdefenders.org): free blue-team labs covering logs, network traffic and forensics.
- **LetsDefend** (letsdefend.io): practice working a SOC alert queue.

Hands-on hacking (legal labs only)

- **TryHackMe** (tryhackme.com): guided, browser-based rooms. Start with the free Pre-Security path.
- **Hack The Box Academy** (academy.hackthebox.com): in-depth modules, with free tiers.
- **PortSwigger Web Security Academy** (portswigger.net/web-security): the best free web-hacking course anywhere.
- **OWASP** (owasp.org): the Top 10, plus the Top 10 for LLM and AI apps (genai.owasp.org).

Competitions & CTFs

- **National Cyber League** (nationalcyberleague.org): student-friendly seasons each spring and fall.
- **picoCTF** (picoctf.org): free, year-round and beginner-friendly, from Carnegie Mellon.
- **CTFtime** (ctftime.org): a calendar of competitions worldwide.

Build a home lab (official downloads only)

- **VirtualBox** (virtualbox.org) or Hyper-V (built into Windows Pro).
- **Windows Server & Windows evaluations:** Microsoft Evaluation Center (microsoft.com/evalcenter).
- **Linux:** Ubuntu (ubuntu.com) and Kali Linux (kali.org) for security tools.
- **Monitoring:** Wazuh (wazuh.com) or Security Onion (securityonionsolutions.com), both free SIEM and monitoring stacks.

Stay current

- **CISA Known Exploited Vulnerabilities catalog** (cisa.gov/known-exploited-vulnerabilities-catalog)
- **SANS Internet Storm Center** (isc.sans.edu): daily notes and a five-minute podcast.
- **Darknet Diaries** (darknetdiaries.com): true stories from the security world.
- **Krebs on Security** (krebsonsecurity.com) and **BleepingComputer** (bleepingcomputer.com): breach and threat news.

Protect yourself first

- **Use a password manager and turn on MFA everywhere.** It's the single best habit in security.
- **Have I Been Pwned** (haveibeenpwned.com): check whether your email has appeared in a breach.
- **EFF Surveillance Self-Defense** (ssd.eff.org): plain-language privacy and security guides.

Communities & events

- **BSides** (securitybsides.com): low-cost local security conferences, and great for meeting people who hire.
- **OWASP chapters, ISSA and ISACA meetups:** local, often free, and full of mentors.
- **My Discord** (discord.gg/WJn4bQV9m9): free 1:1 mentoring, plus workshops and meetings held semi-regularly.

Compiled fall 2026. Links change, so if one breaks, or you know a great free resource I missed, tell me in the Discord. Practice only on systems you own or have permission to test.